

Kubernetes Metrics Discovery

Comprehensive Monitoring Metrics Reference

Overview

This document provides a comprehensive list of Kubernetes metrics used by Last9 for monitoring deployments, pods, and stateful sets. These metrics are essential for effective cluster monitoring, resource management, and troubleshooting.

Total Unique Metrics: 36 metrics

Core Metrics Categories

Deployment Metrics

- kube_deployment_spec_replicas
- kube_deployment_status_replicas_ready
- kube_deployment_status_replicas_updated
- kube_deployment_status_replicas_available
- kube_deployment_status_replicas_unavailable

Pod Metrics

- kube_pod_status_phase
- kube_pod_created
- kube_pod_container_status_running
- kube_pod_container_info
- kube_pod_container_status_restarts_total
- kube_pod_info
- kube_pod_owner
- kube_pod_container_status_waiting_reason
- kube_pod_container_status_terminated
- kube_pod_container_status_last_terminated_reason
- kube_pod_container_status_terminated_reason

StatefulSet Metrics

- kube_statefulset_created
- kube_statefulset_replicas
- kube_statefulset_status_replicas_ready
- kube_statefulset_status_replicas_updated
- kube_statefulset_status_replicas_available
- kube_statefulset_status_replicas_unavailable
- kube_statefulset_spec_replicas
- kube_statefulset_info

Resource Metrics

- kube_pod_container_resource_requests (cpu, memory, ephemeral_storage)
- kube_pod_container_resource_limits (cpu, memory, ephemeral_storage)

Container Metrics

- container_cpu_usage_seconds_total
- container_memory_working_set_bytes
- container_memory_usage_bytes
- container_fs_reads_bytes_total
- container_fs_writes_bytes_total
- container_network_receive_bytes_total
- container_network_transmit_bytes_total
- container_network_receive_errors_total
- container_network_transmit_errors_total
- container_cpu_cfs_throttled_periods_total

Deployment Monitoring

Key Metrics

Metric	Description
kube_deployment_spec_replicas	Desired replicas count
kube_deployment_status_replicas_ready	Ready replicas count
kube_deployment_status_replicas_updated	Updated replicas count
container_cpu_usage_seconds_total	CPU usage (rate)
container_memory_working_set_bytes	Memory usage

Pod Monitoring

Overview Metrics

Metric	Description
kube_pod_status_phase	Pod phase (Running, Pending, etc.)
kube_pod_created	Pod age/creation timestamp
kube_pod_container_status_running	Running containers count
kube_pod_container_status_restarts_total	Container restarts
kube_pod_info	Pod metadata (pod_ip, node, namespace, cluster)

StatefulSet Monitoring

Key Metrics

Metric	Description
kube_statefulset_created	StatefulSet age
kube_statefulset_replicas	Desired replicas
kube_statefulset_status_replicas_ready	Ready replicas
kube_statefulset_info	StatefulSet metadata

Resource Monitoring Categories

CPU Metrics

- kube_pod_container_resource_requests (resource="cpu")
- kube_pod_container_resource_limits (resource="cpu")
- container_cpu_usage_seconds_total (rate)
- container_cpu_cfs_throttled_periods_total

Memory Metrics

- kube_pod_container_resource_requests (resource="memory")
- kube_pod_container_resource_limits (resource="memory")
- container_memory_working_set_bytes
- container_memory_usage_bytes

Storage Metrics

- kube_pod_container_resource_requests (resource="ephemeral_storage")
- kube_pod_container_resource_limits (resource="ephemeral_storage")

Disk I/O Metrics

- container_fs_reads_bytes_total (rate)
- container_fs_writes_bytes_total (rate)

Network Metrics

- container_network_receive_bytes_total (rate)
- container_network_transmit_bytes_total (rate)
- container_network_receive_errors_total (rate)
- container_network_transmit_errors_total (rate)